

YOGESHWARAN G

Software Engineer | Full Stack Developer (MERN)

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SUMMARY

Full-Stack Engineering Lead with 3+ years of experience architecting highly scalable web applications, distributed systems, and dynamic rendering engines. Specialized in Agentic Workflow Orchestration—shifting the traditional "coding lead" paradigm to an AI-augmented architectural supervisor. Recently leveraged agent-native IDEs to architect and deploy an enterprise-grade SaaS product from scratch in just 2 months, establishing strict "human-in-the-loop" guardrails to prevent AI-generated technical debt and ensure system reliability. Proven track record in High-Level Design (HLD) and applying complex Data Structures & Algorithms (DSA) to optimize enterprise platforms.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, HTML5, CSS3

Frontend Architecture: React.js, Redux Toolkit (RTK Query), Material-UI (MUI), State Reconciliation, Dynamic Layout Engines

Backend Architecture: Node.js, Express.js, RESTful APIs, Micro-endpoints

System Design (HLD/LLD): Distributed Systems, Mutex Locking, Message Queues (BullMQ), Event-Driven Architecture, API Gateways

Algorithms (DSA): Recursive Tree Traversal, Dependency Graph Evaluation, RegEx Fuzzy Matching, AST/JSON Path Parsing

Databases & Caching: MongoDB, MySQL, Redis, Schema Normalization, ORM/ODM (Sequelize, Mongoose)

Cloud, DevOps & Security: AWS (Cloud Practitioner), Docker, CI/CD Pipelines, Git, PM2, JWT, OAuth2, 2FA, WebSockets, MQTT

Methodologies & QA: Agile, Scrum, Test-driven Development (TDD), Unit Testing (Jest, Mocha), Test Automation

Generative AI & Tooling: AI-Augmented Systems Design, Agentic Workflows, Systematic Prompt Testing, Rapid Prototyping (Cursor/Copilot)

PROFESSIONAL EXPERIENCE

AICTEC Saudi (A joint venture of Transond Systems) - Coimbatore

March 2024 – Present

Full Stack Developer

Project: WEJHA - Smart Signage Application

- **[AI Orchestration & Architecture]** Spearheaded the architecture and deployment of this enterprise-grade smart signage product from scratch to production in just 2 months, accelerating velocity by 500% via agent-native workflows (Cursor) while establishing strict architectural guardrails to prevent AI-generated technical debt.
- **[DSA & Frontend Architecture]** Engineered an intelligent automated data-mapping engine using recursive tree-traversal algorithms to flatten deeply nested JSON payloads, paired with custom RegEx-powered fuzzy-matching heuristics to automatically bind external API fields to React UI components without manual mapping.
- **[HLD & Scalable Systems]** Architected a horizontally scalable background synchronization engine featuring MongoDB-backed cluster-wide mutex locking, successfully eliminating race conditions and redundant external API polling across multi-instance Node.js deployments.
- **[DSA & Backend Optimization]** Streamlined external data polling by implementing a dependency-graph evaluation algorithm that recursively traverses active schedule logic trees, reducing unnecessary network requests by 75% by updating only actively executing signage triggers.
- **[LLD & Performance]** Built a high-performance React template rendering engine maintaining strict 60FPS timeline animations for heavy multimedia assets, utilizing aggressive state memoization and responsive bounding-box math for pixel-perfect aspect ratio scaling.
- **[HLD & Task Orchestration]** Scaled heavy asynchronous media export pipelines, preventing server CPU exhaustion during peak loads, by orchestrating a Redis-backed BullMQ decentralized task queue with strict worker concurrency isolation and job-state persistence.

Kanmalai Technologies - Coimbatore

December 2022 – February 2024

Full Stack Developer

Project: Digi-Care X - Healthcare Application

- Decoupled the frontend presentation layer from the backend via a normalized Redux state architecture and highly responsive UI components, accelerating patient appointment processing times by 30% and improving overall clinic throughput.
- Enhanced database query execution speeds by 40%, ensuring sub-second data retrieval for complex Electronic Health Records (EHR), by restructuring legacy MySQL schemas and implementing strategic indexing through the ORM layer.
- Reduced patient no-show rates by 15%, as measured by quarterly attendance metrics, by engineering an event-driven notification service using Firebase Cloud Messaging and SMS gateways for real-time appointment lifecycle alerts.

Project: Employee Information and Leave Management System (HRMS)

- Decreased system response latency for core HR operations by 40%, supporting concurrent multi-tenant access, by refactoring monolithic Express.js controllers into highly cohesive, async-optimized RESTful micro-endpoints.
- Refined payroll processing efficiency by 30%, saving the HR department significant manual calculation hours, by automating the attendance aggregation data pipeline and ensuring strict chronological data consistency.

10X Academy - Hyderabad

June 2022 – November 2022

MERN Stack Intern

Project: Real Estate Application

- Increased active property listing engagement by 30%, by engineering a highly responsive, mobile-first React property discovery platform leveraging debounced search inputs and optimized component rendering.
- Reduced listing search latency by 50% and query response times by 40%, by designing denormalized MongoDB schemas tailored for complex, multi-criteria geospatial and price-range aggregations.

EDUCATION

- **BE in Mechanical Engineering**, Bannari Amman Institute of Technology - Sathyamangalam (2016-2020) - 67%
- **HSC**, RKR GRKS Matric Higher Secondary School - Udumalpet (2014-2016) - 94%
- **SSLC**, RKR GRKS Matric Higher Secondary School - Udumalpet (2014) - 94%

AWARDS & CERTIFICATIONS

- Recipient of the Bright Debutant Award for outstanding performance.
- AWS Certified Cloud Practitioner